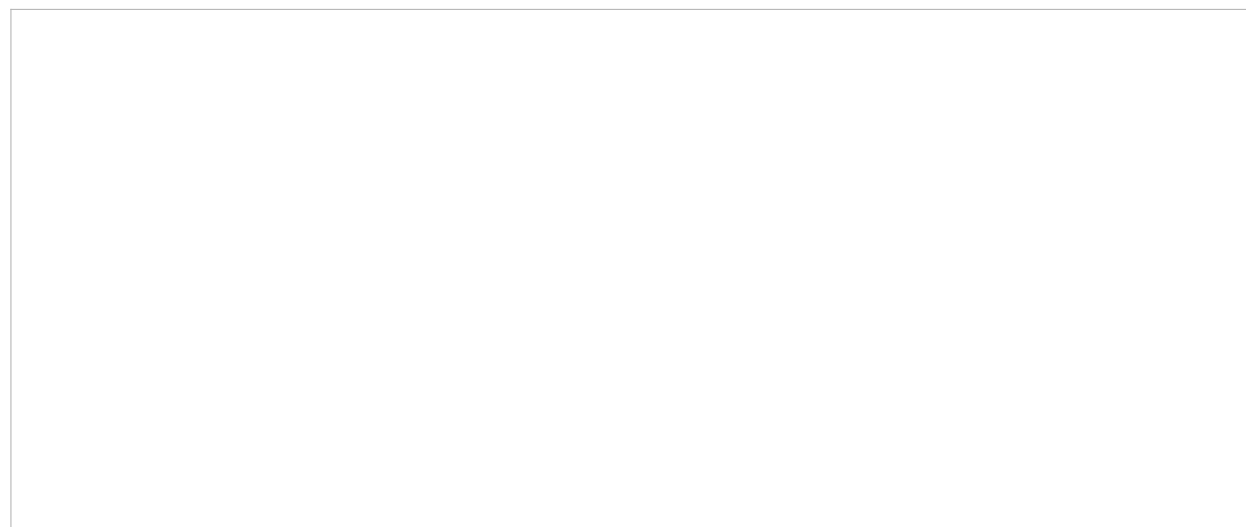


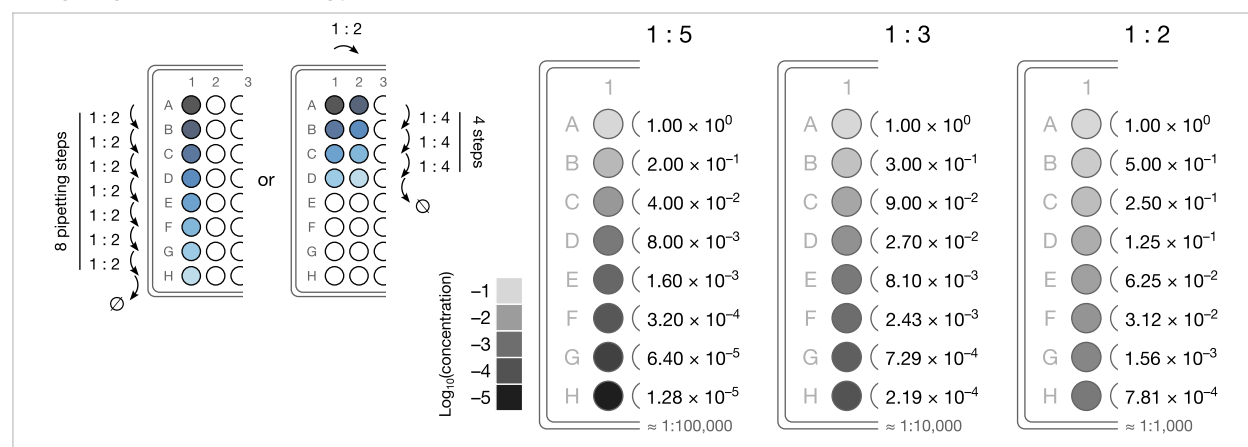
Resources

Accurate volume transfer and pipetting

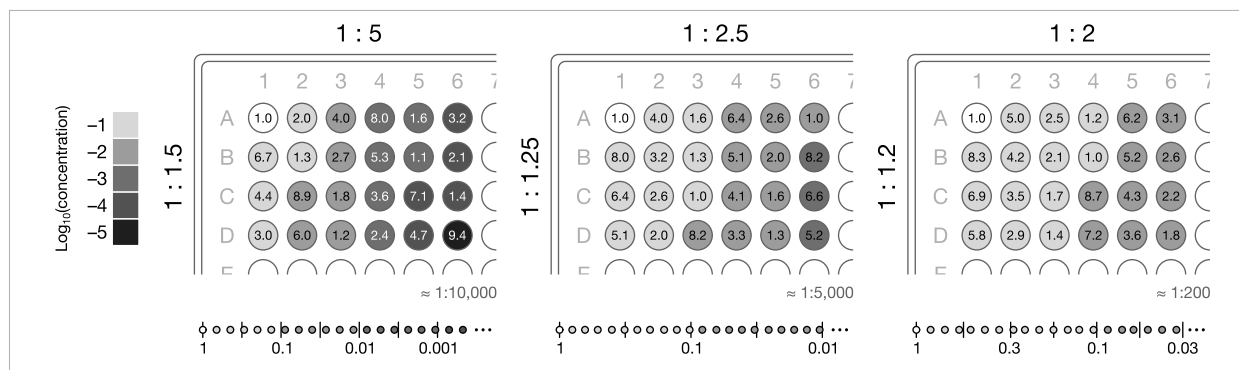


In Step 3: Pipetting motion during aspirating and dispensing. Modified from Gilson PIPETMAN® User's Guide Pub. No. LT801122/E.

Designing a dilution strategy

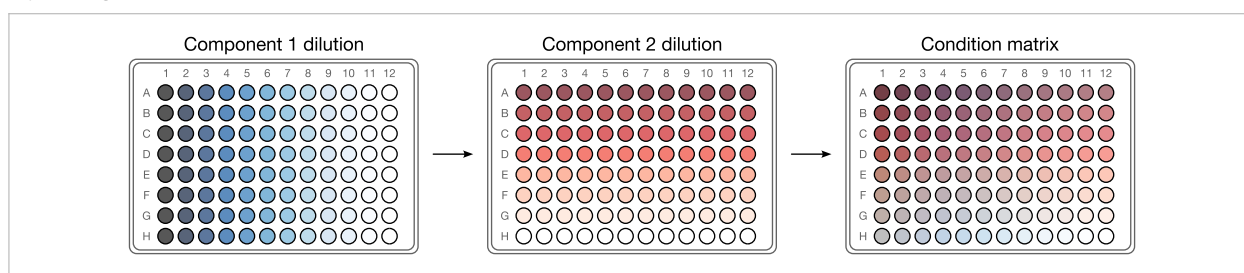


In Step 2: Even logarithmic serial dilutions with constant dilution factors of 1 : 5, 1 : 3, and 1 : 2 in a 1D layout. Each step reduces the concentration by the stated factor. Larger dilution factors cover more orders of magnitude in fewer steps but with coarser spacing between concentrations. If a multichannel pipette is available, pipetting steps can be reduced.



In Step 2: Combining two dilution factors across rows and columns produces finer logarithmic, almost linear spacing than either factor alone.

Pipetting a dilution series



In Step 8: Pipetting workflow for a two-component 2D dilution. Component 1 is serially diluted across columns; component 2 is independently diluted across rows. If enough excess is prepared, the second dilution can be applied within the same plate. The resulting matrix covers a wide combinatorial range of conditions, enabling the analysis of synergy, antagonism, or dose-response effects between drugs, ligands, enzymes, or cells.

Change log

2021-12-20	Benjamin C. Buchmuller	Adaptation as SOP.
2023-11-22	Benjamin C. Buchmuller	Reformatting.
2025-04-24	Benjamin C. Buchmuller	Added guidance on pipetting and two-dimensional serial dilutions.
2026-02-26	Benjamin C. Buchmuller	Integrate resource table into procedure.

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